Topic - 11

Polymorphism:

Polymorphism means "many forms". It allows the same method name (or) function to behave differently depending on the object (or) input..

Polymorphism : same name, different behavior..

- one Interface
- multiple implementation.

Real world Example:

Think about a remote control button.

You press the same buttons

- Tv → changes channel
- Ac → changes temperature
- Music system → changes song.

Same action → different behavior..

Polymorphism

Method overriding.
(Runtime polymorphism).

Duck Typing
(Python) Mostly
very Important)

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① Method overriding:-

```
class Animal:
    def sound(self):
        Print("Animal makes sound")

class Dog(Animal):
    def sound(self):
        Print("Dog barks")

class cat(Animal):
    def sound(self):
        Print("cat meows")

d = Dog()
c = cat()
d.sound() # Dog barks
c.sound() # cat meow.
```

All classes have same method name + sound()

But behavior is different

This is runtime

Polymorphism

② Duck Typing:

If an object behaves like a certain type, python doesn't care if

class

"If it look like a duck and behave like a duck, it's a duck."

```
class Dog:
    def sneak(self):
        Print("Bark")

class cat:
    def sneak(self):
        Print("Meow")

def make_sound(animal):
    animal.sneak()

d = Dog()
c = cat()

make_sound(d)
make_sound(c)
```

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Polymorphism with function

```
def add(a,b):
    return a+b

print(add(2,3)) # 5
print(add("HI","HELLO") # HI HELLO
```

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